

Assignment 5-6

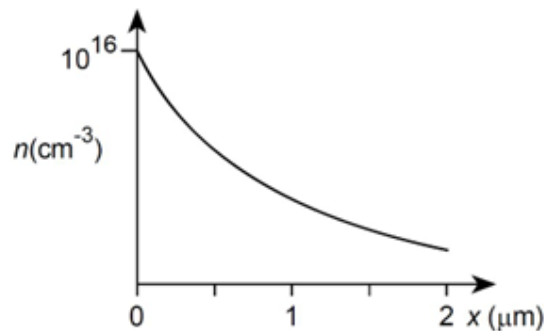
Doped Si and PN Junction

(Due on **May. 21, 2026**)

Question 1

An n-type semiconductor sample is non-uniformly doped, as a result of which the electron concentration $n(x)$ over a section of sample from $0 \leq x \leq 2x_0$ is given by $n(x) = n_1 \exp(-x/x_0)$, where $n_1 = 10^{16} \text{ cm}^{-3}$, and $x_0 = 1 \text{ } \mu\text{m}$, as shown.

- The sample is at thermal equilibrium. Does an electric field exist in this sample? Discuss. (6 pts)
- By considering the drift and diffusion current components of the electron current, derive an expression for the electric field $E(x)$. Sketch the field as a function of x . Calculate the value of the electric field, and note its direction. (12 pts)
- By integrating the electric field, sketch the potential distribution along the sample. The electric field is related to the potential $V(x)$ by $E = -dV/dx$. Work out the potential difference over this section ($0 \leq x \leq 2x_0$). (12 pts)
- The vacuum level is related to the potential by $E_{vac} = -eV(x)$. Sketch the energy band diagram for the sample assuming an electron affinity of 4.05 eV for the silicon. (5 pts)
- Obtain an expression for $E_C(x) - E_F$. Note that the Fermi level is uniform throughout in this case since the sample is at thermal equilibrium. Comment. (5 pts)



(a) The sample is at thermal equilibrium. Does an electric field exist in this sample? Discuss.

At thermal equilibrium, there is not net current flow, and hence, $J = J_n + J_p = 0$. However, we can further state that $J_n = J_p = 0$ by considering what would happen if this were not the case, i.e., $J_n = -J_p$. This would imply that electrons and holes would move in the same direction in the sample, and in steady-state, electron-hole pair generation (requiring energy input) at one end of the sample, and electron-hole pair annihilation (giving out energy) at the other end of the sample, which is not possible.

Due to non-uniform doping, there is a diffusion of electrons from the higher concentration region to the lower concentration region. This tendency towards diffusion is counterbalanced by an opposite force on the electrons due the electric field established by the net charge in the sample resulting from the diffusion of the electrons. The fixed ionized dopants in the background do not quite balance the electron distribution and results in space-charge in the sample.

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(b) By considering the drift and diffusion current components of the electron current, derive an expression for the electric field $E(x)$. Sketch the field as a function of x . Calculate the value of the electric field, and note its direction. [Ans: 258.6 V/cm]

Since $J_n = 0$, then

$$J_{n,diff} + J_{n,drift} = 0$$

$$eD_n \frac{dn}{dx} + e\mu_n nE = 0 \Rightarrow E = -\frac{D_n}{\mu_n} \frac{1}{n} \frac{dn}{dx} = -\frac{kT}{e} \frac{1}{n} \frac{dn}{dx}$$

using the Einstein relationship. We have $n(x) = n_1 \exp(-x/x_0)$, so

$$E = -\frac{kT}{e} \left(\frac{1}{n_1 \exp(-x/x_0)} \right) \left(-\frac{n_1}{x_0} \exp(-x/x_0) \right) = \frac{kT}{e} \frac{1}{x_0}$$

i.e., the electric field is constant for this exponential distribution of electrons. Since $kT/e = 0.02586$ V and $x_0 = 10^{-4}$ cm, the electric field is 258.6 V/cm. Note that the electric field is in the +ve x -direction, which means that the electrons drift in the -ve x -direction to counter the diffusion of the electrons in the +ve x -direction, which is correct.

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(c) By integrating the electric field, sketch the potential distribution along the sample. The electric field is related to the potential $V(x)$ by $E = -dV/dx$. Work out the potential difference over this section ($0 \leq x \leq 2x_0$). [Ans: -0.0571V]

The potential / voltage along the sample can be obtained by integrating the (constant) electric field. Hence

$$V(x) - V(0) = -\int_0^x E \cdot dx = -258.6 \cdot x$$

The potential difference over the section of the sample is

$$V(2x_0) - V(0) = -258.6 \cdot 2 \times 10^{-4} = -0.05172 \text{ V}$$

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(d) The vacuum level is related to the potential by $E_{vac} = -eV(x)$. Sketch the energy band diagram for the sample assuming an electron affinity of 4.05 eV for the silicon.

(Sketch on the next page.) The vacuum level is related to the potential in the bar by $E_{vac} = -eV(x)$, and the conduction and valence band edges are parallel to this. As the Fermi level is uniform throughout the sample (since it is at thermal equilibrium), the conduction band edge is seen to slope up from the left hand side to the right hand side. On the left hand side, the conduction band edge is closer to the Fermi level and the electron concentration is higher, which ties in with the given electron distribution. Electrons will thus diffuse to the right-hand side. On the other hand, electrons are seen to drift from higher energy levels near the conduction band edge on the right hand side, to lower energy levels on the left hand side.

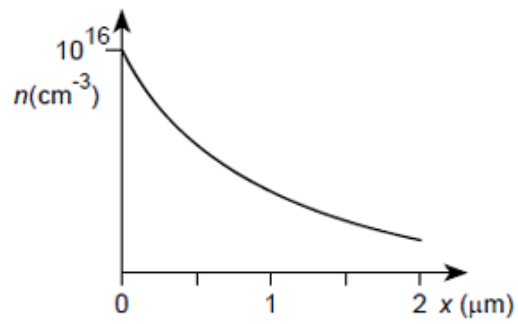
(e) Obtain an expression for $E_C(x) - E_F$. Note that the Fermi level is uniform throughout in this case since the sample is at thermal equilibrium. Comment.

Two ways of doing this. Most direct way is from the electron concentration expression.

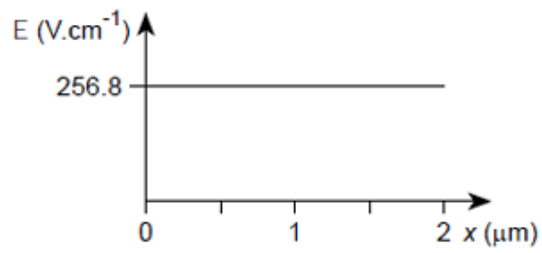
$$\begin{aligned}n(x) &= N_C \exp\left(-\frac{E_C(x) - E_F}{kT}\right) \\ \Rightarrow E_C(x) - E_F &= kT \ln\left(\frac{N_C}{n(x)}\right) \\ &= kT \ln N_C - kT \ln\left\{n_1 e^{-x/x_0}\right\} \\ &= kT \ln\left(\frac{N_C}{n_1}\right) + kT \frac{x}{x_0}\end{aligned}$$

Note that the vacuum level (which is parallel to the conduction band edge) is the mirror image of the potential (with a scaling factor of e).

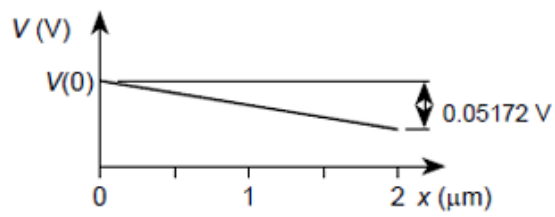
The alternative approach is from the energy band diagram in part (d) which gives the same answer (and incidentally one obtains the Einstein Relationship, $D/\mu = kT/e$).



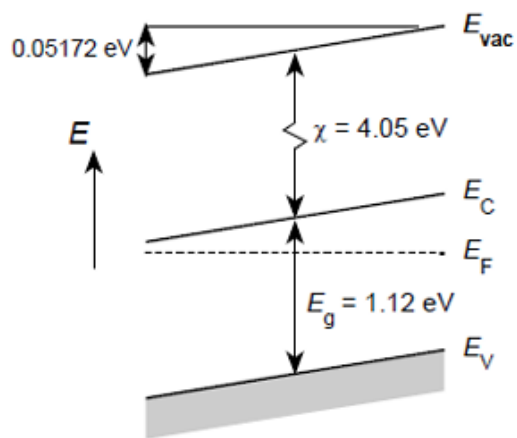
Electron Concentration



Electric Field



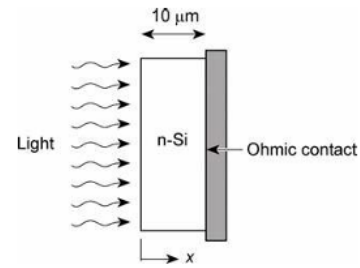
Potential



Energy band diagram
(NOT TO SCALE)

Question 2

A photodetector structure consists of a silicon film of $10\text{ }\mu\text{m}$ thickness on an ohmic contact, as shown. The silicon is uniformly doped n-type. Short wavelength light illuminates the surface (at $x = 0$) continuously and generates electron-hole pairs only at the surface. As a result, the excess minority carrier concentration at the surface is found to be 10^{14} cm^{-3} . At the ohmic contact, the carrier concentrations are at their thermal equilibrium values. Use the following data given for the silicon which is at 300 K:



Intrinsic carrier concentration: $6.73 \times 10^9\text{ cm}^{-3}$

Electron mobility: $1500\text{ cm}^2\text{V}^{-1}\text{s}^{-1}$

Minority carrier lifetime: $20\text{ }\mu\text{s}$

Resistivity: $3\text{ }\Omega\text{-cm}$

Hole mobility: $450\text{ cm}^2\text{V}^{-1}\text{s}^{-1}$

- (a) Determine the thermal equilibrium carrier concentrations; (5 pts)
- (b) Determine the minority carrier diffusion length; (5 pts)
- (c) Sketch the minority carrier concentration as a function of distance x ; (5 pts)
- (d) Determine the diffusion current density of the minority carriers at the surface, stating the direction of the current density. (5 pts)

(a) Determine the thermal equilibrium carrier concentrations; [Ans: 1.39×10^{15} , 3.27×10^4]

Since we are told that the silicon is n-type, the majority carrier concentration can be derived from the resistivity:

$$\rho = \frac{1}{\sigma} = \frac{1}{ne\mu_n}$$

$$n = \frac{1}{\rho e \mu_n} = \frac{1}{3 \cdot 1.602 \times 10^{-19} \cdot 1500} = 1.387 \times 10^{15} \text{ cm}^{-3}$$

$$p = \frac{n_i^2}{n} = \frac{(6.73 \times 10^9)^2}{1.387 \times 10^{15}} = 3.265 \times 10^4 \text{ cm}^{-3}$$

(b) Determine the minority carrier diffusion length; [Ans: $152.5 \mu\text{m}$]

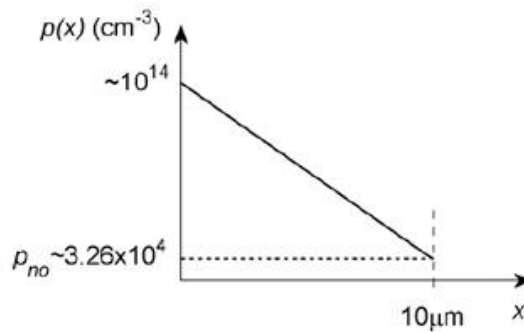
$$L_p = \sqrt{D_p \tau_p}$$

$$\frac{D_p}{\mu_p} = \frac{kT}{e} \Rightarrow D_p = 0.02586 \cdot 450 = 11.637 \text{ cm}^2 \text{ s}^{-1}$$

$$\therefore L_p = \sqrt{11.637 \cdot 20 \times 10^{-6}} = 1.526 \times 10^{-2} \text{ cm} = 152.5 \mu\text{m}$$

(c) Sketch the minority carrier concentration as a function of distance x :

Since the thickness of the film is much less than the minority carrier diffusion length, the excess carrier concentration will drop linearly to zero at the ohmic contact.



$$np = n_i^2$$

$$p_{no} = \frac{n_i^2}{n_{no}} = \frac{(6.73 \times 10^9)^2}{1.39 \times 10^{15}} = 3.26 \times 10^4$$

(d) Determine the diffusion current density of the minority carriers at the surface, stating the direction of the current density. [Ans: 0.186 A/cm^2]

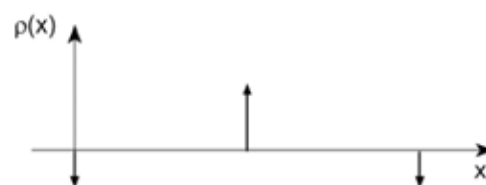
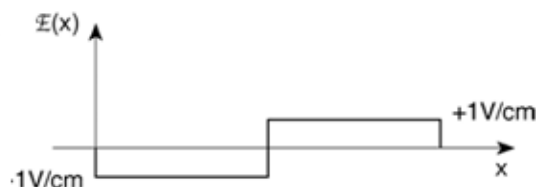
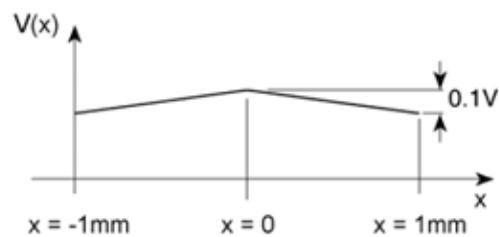
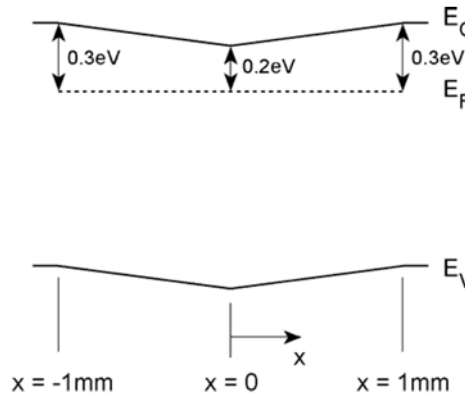
The hole diffusion current density is given by:

$$J_{p, \text{diff}} = -eD_p \frac{dp}{dx} = -e \cdot 11.637 \cdot \frac{(3.26 \times 10^4 - 10^{14})}{10 \times 10^{-4}} = 0.186 \text{ Acm}^{-2}$$

Current flow is from left to right (positive for increasing x).

Question 3:

The figure on the right shows the energy-band diagram of part of a bar of silicon at thermal equilibrium at 300K. Sketch the potential, electric field, and charge density along the length of the bar. Indicate the E-field values. (15 pts)



$$E_{vac} = -e\phi(x)$$

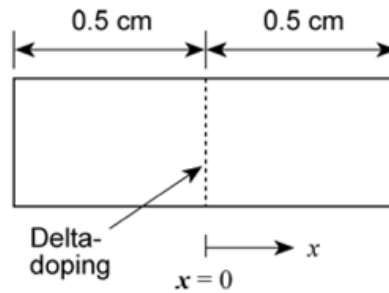
$$E(x) = -\frac{d\phi(x)}{dx}$$

$$\frac{dE(x)}{dx} = \frac{\rho(x)}{\epsilon_s}$$

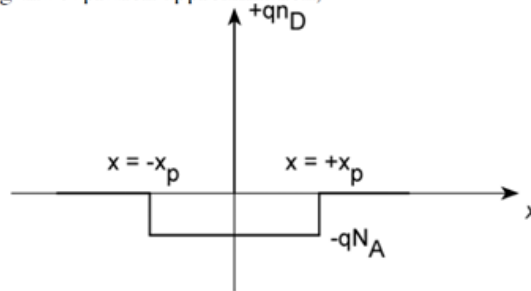
Question 4

The figure below shows a 1 cm length of silicon doped with acceptors at a concentration of 10^{13} cm^{-3} . The doping is uniform except at $x = 0$ where there is an additional sheet doping (delta-doping) of donors at a density of 10^{10} cm^{-2} (i.e., total of 10^{10} donors per cm^2 cross-sectional area). You may assume that all the dopants are ionized.

- Sketch the charge density along the length of the silicon sample. Using the depletion approximation, determine the total width of the depletion region. (5 pts)
- Determine the maximum electric field (in units of V/cm) in the silicon. Sketch the electric field and potential along the length of the silicon. (10 pts)
- Determine the position of the Fermi level relative to the conduction band edge at $x = 0$. (5 pts)
- Sketch the energy band diagram showing the band edges, the Fermi level, and the vacuum level. (5 pts)



From neutrality, the total space-charge must cancel out. Since all the donors and acceptors are ionized, we have, using the depletion approximation,



Hence

$$2qN_A x_p = qn_D$$

$$\Rightarrow x_p = \frac{n_D}{2N_A} = \frac{10^{10}}{2 \cdot 10^{13}} = 5 \times 10^{-4} \text{ cm}.$$

(b) Electric field

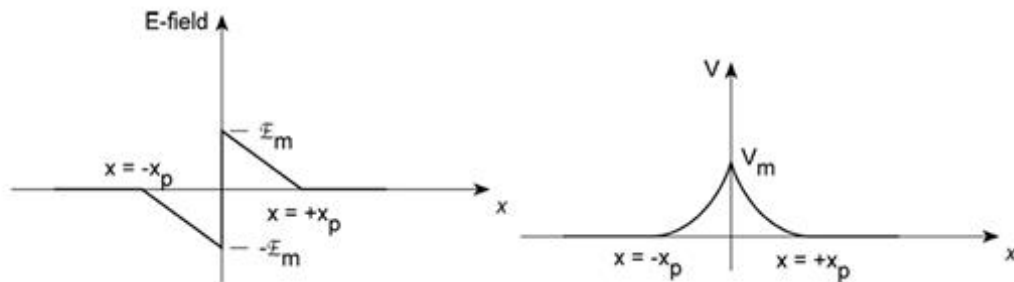
The electric field is obtained by integrating the charge density. From Poisson's equation, we have,

$$\frac{dE}{dx} = \frac{\rho}{\epsilon_0 \epsilon_r} \Rightarrow E = \int \frac{\rho}{\epsilon_0 \epsilon_r} dx$$

The magnitude of the peak electric field at $x = 0$ is

$$|E_m| = \frac{q N_A x_p}{\epsilon_0 \epsilon_r} = \frac{e \cdot 10^{13} \cdot 5 \times 10^{-4}}{\epsilon_0 \cdot 11.7} = 773 \text{ V/cm}$$

Sketch of electric field



The potential in the bar is obtained by integrating the E-field, viz. $V = -\int E dx$

(c) Position of Fermi level

We need to first work out the maximum potential at $x = 0$ in the bar as this represents the "dip" in the energy band at $x = 0$. Integrating the electric field from $x = -x_p$ to $x = 0$, or simply the area under the E-field curve, we get,

$$V_m = -\left(-\frac{1}{2} E_m x_p\right) = \frac{1}{2} \cdot 773 \cdot 5 \times 10^{-4} = 0.193 \text{ V}$$

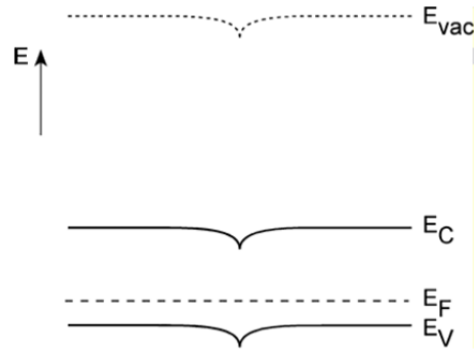
The position of the Fermi level relative to the conduction-band edge in the neutral p regions can be obtained from,

$$\begin{aligned} E_C - E_F &= kT \ln\left(\frac{N_C}{n}\right) \\ n &= \frac{n_i^2}{p} = \frac{n_i^2}{N_A} = \frac{(6.95 \times 10^9)^2}{10^{13}} = 4.83 \times 10^6 \text{ cm}^{-3} \\ \therefore E_C - E_F &= 0.0259 \times \ln \frac{2.8 \times 10^{19}}{4.83 \times 10^6} = 0.761 \text{ eV} \end{aligned}$$

Finally, $E_C - E_F$ at $x = 0$ is

$$0.761 - eV_m = 0.761 - 0.193 = 0.568 \text{ eV}$$

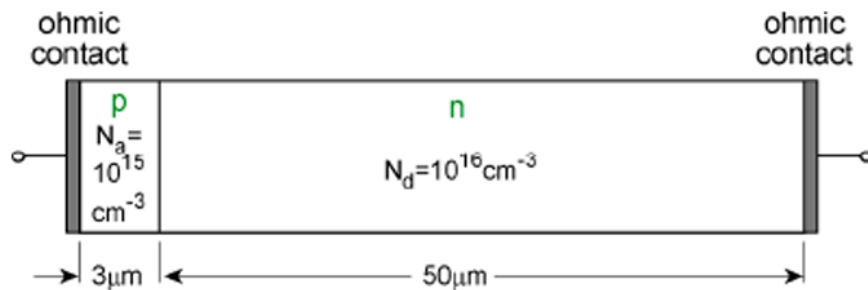
(d) Sketch of energy band-diagram



Question 5

An abrupt Si PN-junction diode (shown below) with cross-sectional area of $2 \times 10^{-5} \text{ cm}^2$ is reverse-biased by 2 V. The minority carrier lifetimes on both sides of the junction are 10^{-7} s . Assume $T = 300 \text{ K}$, electron mobility $\mu_n = 1300 \text{ cm}^2/\text{V}\cdot\text{s}$, hole mobility $\mu_p = 400 \text{ cm}^2/\text{V}\cdot\text{s}$.

- Calculate the depletion widths on both sides of the junction. (5 pts)
- Determine the minority carrier diffusion lengths. (5 pts)
- Calculate the current flowing through the device, assuming that breakdown has not occurred. (5 pts)
- Calculate the value of the junction capacitance, and estimate the series resistance of the device. (5 pts)
- Propose a small-signal equivalent circuit model of the junction at this bias condition, and give the values of the circuit components. (5 pts)
- If the reverse bias is increased to 4 V, would the current flowing through the device change? How would this affect your small-signal equivalent model of part (e)? Discuss. (10 pts)



(a)

$$V_{bi} = V_t \ln \left(\frac{N_a N_d}{n_i^2} \right) = 0.0259V \times \ln \left(\frac{10^{15} \text{ cm}^{-3} \times 10^{16} \text{ cm}^{-3}}{(6.95 \times 10^9 \text{ cm}^{-3})^2} \right) = 0.675V$$

$$V_{total} = V_{bi} + V_R = 0.675V + 2V = 2.675V$$

$$W = \left[\frac{2\epsilon_s V_{total}}{e} \left(\frac{N_a + N_d}{N_a \cdot N_d} \right) \right]^{1/2}$$

$$= \left[\frac{2 \cdot (11.7 \times 8.854 \times 10^{-14} \text{ F/cm}) \cdot 2.675V}{1.6 \times 10^{-19} \text{ C}} \left(\frac{10^{15} \text{ cm}^{-3} + 10^{16} \text{ cm}^{-3}}{10^{15} \text{ cm}^{-3} \times 10^{16} \text{ cm}^{-3}} \right) \right]^{1/2}$$

$$= 1.95 \times 10^{-4} \text{ cm} = 1.95 \mu\text{m}$$

$$x_p = \frac{N_d}{N_a + N_d} \cdot W = \frac{10^{16} \text{ cm}^{-3}}{10^{15} \text{ cm}^{-3} + 10^{16} \text{ cm}^{-3}} \times 1.94 \mu\text{m} = 1.77 \mu\text{m}$$

$$x_n = \frac{N_a}{N_a + N_d} \cdot W = \frac{10^{15} \text{ cm}^{-3}}{10^{15} \text{ cm}^{-3} + 10^{16} \text{ cm}^{-3}} \times 1.94 \mu\text{m} = 0.18 \mu\text{m}$$

(b)

$$\frac{D_n}{\mu_n} = \frac{kT}{e} \Rightarrow D_n = 1300 \text{ cm}^2/\text{V} \cdot \text{s} \times 0.0259V = 33.67 \text{ cm}^2/\text{s}$$

$$L_n = \sqrt{D_n \tau_n} = \sqrt{33.67 \text{ cm}^2/\text{s} \times 10^{-7} \text{ s}} = 1.8 \times 10^{-3} \text{ cm} = 18 \mu\text{m}$$

(c)

$$\frac{D_p}{\mu_p} = \frac{kT}{e} \Rightarrow D_p = 400 \text{ cm}^2/\text{V} \cdot \text{s} \times 0.0259V = 10.36 \text{ cm}^2/\text{s}$$

$$L_p = \sqrt{D_p \tau_p} = \sqrt{10.36 \text{ cm}^2/\text{s} \times 10^{-7} \text{ s}} = 1 \times 10^{-3} \text{ cm} = 10 \mu\text{m}$$

$$W_p = 3 \mu\text{m} - 1.76 \mu\text{m} = 1.24 \mu\text{m}$$

$$W_n = 50 \mu\text{m} - 0.18 \mu\text{m} = 49.82 \mu\text{m}$$

$$n_{p0} = \frac{n_i^2}{N_a} = \frac{(6.95 \times 10^9 \text{ cm}^{-3})^2}{10^{15} \text{ cm}^{-3}} = 4.83 \times 10^4 \text{ cm}^{-3}$$

$$p_{n0} = \frac{n_i^2}{N_d} = \frac{(6.95 \times 10^9 \text{ cm}^{-3})^2}{10^{16} \text{ cm}^{-3}} = 4.83 \times 10^3 \text{ cm}^{-3}$$

So we have $W_p \ll L_n$.

$$J_{total} = J_{n,diff} + J_{p,diff} = \left(\frac{e D_n n_{p0}}{W_p} + \frac{e D_p p_{n0}}{L_p} \right) \cdot \left(e^{\frac{V_a}{V_t}} - 1 \right) \approx - \left(\frac{e D_n n_{p0}}{W_p} + \frac{e D_p p_{n0}}{L_p} \right) = -J_s$$

$$= - \left(\frac{1.6 \times 10^{-19} \text{ C} \times 33.67 \text{ cm}^2/\text{s} \times 4.83 \times 10^4 \text{ cm}^{-3}}{1.24 \mu\text{m}} + \frac{1.6 \times 10^{-19} \text{ C} \times 10.36 \text{ cm}^2/\text{s} \times 4.83 \times 10^3 \text{ cm}^{-3}}{10 \mu\text{m}} \right) = -2.12 \times 10^{-9} \text{ A/cm}^2$$

Finally,

$$I = J_{total} \times S = -2.12 \times 10^{-9} \text{ A/cm}^2 \times 2 \times 10^{-5} \text{ cm}^2 = -4.24 \times 10^{-14} \text{ A}$$

(d)

$$C_j' = \frac{\epsilon_s}{W} = \frac{11.7 \times 8.854 \times 10^{-14} \text{ F/cm}}{1.94 \times 10^{-4} \text{ cm}} = 5.34 \times 10^{-9} \text{ F/cm}^2$$

$$C_j = C_j' \times S = 5.34 \times 10^{-9} \text{ F/cm}^2 \times 2 \times 10^{-5} \text{ cm}^2 = 1.07 \times 10^{-13} \text{ F}$$

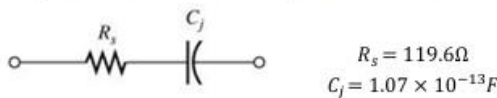
In the n-neutral region, the current is entirely carried by the majority carriers (electrons) for most of the neutral region, but a fraction is carried by minority carriers (holes) diffusing near the SCR. This latter is small since $J_{s,p} \ll J_s$. So we can consider the length of the n-neutral region as a resistor with conductivity:

$$r_s = r_{sn} \approx \frac{W_n}{en_0\mu_n A}$$

$$= \frac{50-0.18 \mu\text{m}}{1.6 \times 10^{-19} \text{ C} \times 10^{16} \text{ cm}^{-3} \times 1300 \text{ cm}^2/\text{V} \cdot \text{s} \times 2 \times 10^{-5} \text{ cm}^2} = 119.6 \Omega$$

In the p-region, most of the current is carried by the minority electrons which diffuse to the ohmic contact with minimal recombination. The current is thus mainly diffusion current. The majority hole current is small (we already observed that $J_{s,p} \ll J_s$), so there is negligible E-field in the neutral p region, and there is insignificant potential drop across the $1.6 \mu\text{m}$ neutral region.

(e) Under reverse bias of more than a few kT/e , the current normally saturates at the reverse saturation current. Under such circumstances, $dI/dV = g_d = 0$, and we only have the junction capacitance in series with the equivalent series resistance worked out earlier.



*(f) If the reverse bias is increased to 4V, would the current flowing through the device change? How would this affect your small-signal equivalent model of part (e)? Discuss. For discussion if time permits. Tutors: some of the concepts introduced here will re-appear in the BJT part. Here we see the same effect that leads to base width modulation & the Early effect in the BJT as V_{BC} changes.

If the reverse bias increases, the width of the space charge region increases, with most of the increase taking place on the lighter-doped p-side. We can calculate the depletion widths as above, giving:

$$W = \left\{ \frac{2 \cdot 8.854 \times 10^{-14} \cdot 11.7(4 + 0.675)}{e} \left[\frac{10^{15} + 10^{16}}{10^{15} \cdot 10^{16}} \right] \right\}^{1/2} = 2.58 \times 10^{-4} \text{ cm} = 2.58 \mu\text{m}$$

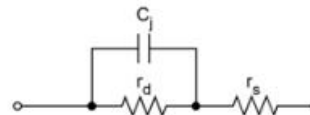
$$x_p = 2.58 \times \frac{10^{16}}{10^{15} + 10^{16}} = 2.34 \mu\text{m} \quad ; \quad x_n = W - x_p = 0.24 \mu\text{m}$$

The width of the neutral p-region is now $(3-2.34) = 0.66 \mu\text{m}$. This means that the gradient of the excess electron concentration is now higher, leading to a higher $J_{s,n}$ and J_s overall:

$$J_s = J_{s,n} + J_{s,p} = \frac{eD_n n_{p0}}{W_p} + \frac{eD_p p_{n0}}{L_p}$$

$$= \frac{e \cdot 33.7 \cdot 4.83 \times 10^4}{0.66 \times 10^{-4}} + \frac{e \cdot 10.36 \cdot 4.83 \times 10^3}{10.2 \times 10^{-4}}$$

$$= 3.95 \times 10^{-9} + 7.9 \times 10^{-12} = 3.96 \times 10^{-9} \text{ A/cm}^2$$



Note that the current has increased significantly with an increase in reverse bias. On the I-V curve, the reverse saturation current thus increases with more negative V. In terms of an equivalent circuit model, this can be modelled by a finite equivalent diode resistance r_d . In a long-base diode, J_s does not increase, as the shape of the minority carrier distributions does not change as the depletion region widens.

Question 6

Consider an abrupt Si PN-junction diode at 300 K. The doping concentrations are $10^{17}/\text{cm}^3$ acceptors on the p-side, and $10^{18}/\text{cm}^3$ on the n-side. The minority carrier lifetimes are 500 ns and 2 ns, on the p- and n- sides, respectively. The length of the neutral regions is 100 μm for both sides. The cross-sectional area is 1 mm^2 .

- (a) Is the diode long- or short-base on (i) the p-side, and (ii) the n-side? (10 pts)
- (b) Calculate the current through the diode when the diode is forward-biased at 0.6V. (5 pts)
- (c) As a fraction of the total current, what is the proportion of the electron current flowing through the space-charge region? (5 pts)
- (d) How would you increase the proportion of electron current of part (c) even further? What are the implications of your approach? (5 pts)

(a)

$$\begin{aligned}
V_{bi} &= V_t \ln \left(\frac{N_a N_d}{n_i^2} \right) = 0.0259V \times \ln \left(\frac{10^{17} \text{ cm}^{-3} \times 10^{18} \text{ cm}^{-3}}{(6.95 \times 10^9 \text{ cm}^{-3})^2} \right) = 0.91V \\
W &= \left[\frac{2\epsilon_s V_{bi}}{e} \left(\frac{N_a + N_d}{N_a \cdot N_d} \right) \right]^{1/2} \\
&= \left[\frac{2 \cdot (11.7 \times 8.854 \times 10^{-14} \text{ F/cm}) \cdot 0.91V}{1.6 \times 10^{-19} \text{ C}} \left(\frac{10^{17} \text{ cm}^{-3} + 10^{18} \text{ cm}^{-3}}{10^{17} \text{ cm}^{-3} \times 10^{18} \text{ cm}^{-3}} \right) \right]^{1/2} \\
&= 1.14 \times 10^{-5} \text{ cm} = 0.114 \mu\text{m} \\
x_p &= \frac{N_d}{N_a + N_d} \cdot W = \frac{10^{18} \text{ cm}^{-3}}{10^{17} \text{ cm}^{-3} + 10^{18} \text{ cm}^{-3}} \times 0.114 \mu\text{m} = 0.104 \mu\text{m} \\
x_n &= \frac{N_a}{N_a + N_d} \cdot W = \frac{10^{17} \text{ cm}^{-3}}{10^{17} \text{ cm}^{-3} + 10^{18} \text{ cm}^{-3}} \times 0.114 \mu\text{m} = 0.010 \mu\text{m} \\
\frac{D_n}{\mu_n} &= \frac{kT}{e} \Rightarrow D_n = 750 \text{ cm}^2/\text{V} \cdot \text{s} \times 0.0259V = 19.425 \text{ cm}^2/\text{s} \\
L_n &= \sqrt{D_n \tau_n} = \sqrt{19.425 \text{ cm}^2/\text{s} \times 5 \times 10^{-7} \text{ s}} = 3.12 \times 10^{-3} \text{ cm} = 31.2 \mu\text{m} \\
\frac{D_p}{\mu_p} &= \frac{kT}{e} \Rightarrow D_p = 150 \text{ cm}^2/\text{V} \cdot \text{s} \times 0.0259V = 3.885 \text{ cm}^2/\text{s} \\
L_p &= \sqrt{D_p \tau_p} = \sqrt{3.885 \text{ cm}^2/\text{s} \times 2 \times 10^{-9} \text{ s}} = 8.81 \times 10^{-5} \text{ cm} = 0.88 \mu\text{m} \\
W_p &= 100 \mu\text{m} - 0.104 \mu\text{m} = 99.896 \mu\text{m} \\
W_n &= 100 \mu\text{m} - 0.010 \mu\text{m} = 99.990 \mu\text{m}
\end{aligned}$$

We have $W_p > L_n$ and $W_n > L_p$, so the diode is long-base both on the p-side and the n-side.

(b)

$$\begin{aligned}
n_{p0} &= \frac{n_i^2}{N_a} = \frac{(6.95 \times 10^9 \text{ cm}^{-3})^2}{10^{17} \text{ cm}^{-3}} = 483 \text{ cm}^{-3} \\
p_{n0} &= \frac{n_i^2}{N_d} = \frac{(6.95 \times 10^9 \text{ cm}^{-3})^2}{10^{18} \text{ cm}^{-3}} = 48.3 \text{ cm}^{-3} \\
J_s &= \frac{eD_n n_{p0}}{L_n} + \frac{eD_p p_{n0}}{L_p} \\
&= \frac{1.6 \times 10^{-19} \text{ C} \times 19.425 \text{ cm}^2/\text{s} \times 483 \text{ cm}^{-3}}{31.2 \mu\text{m}} \\
&\quad + \frac{1.6 \times 10^{-19} \text{ C} \times 3.885 \text{ cm}^2/\text{s} \times 48.3 \text{ cm}^{-3}}{0.88 \mu\text{m}} = 8.22 \times 10^{-13} \text{ A/cm}^2 \\
J &\approx J_s \cdot e^{V_a/V_t} = 8.22 \times 10^{-13} \text{ A/cm}^2 \times e^{0.6V/0.0259V} = 0.0095 \text{ A/cm}^2 \\
I &= J_s \cdot A = 0.0095 \text{ A/cm}^2 \times 1 \text{ mm}^2 = 9.5 \times 10^{-5} \text{ A}
\end{aligned}$$

(c)

$$\frac{\frac{eD_n n_{p0}}{L_n}}{\frac{eD_n n_{p0}}{L_n} + \frac{eD_p p_{n0}}{L_p}}$$

$$= \frac{\frac{1.6 \times 10^{-19} C \times 19.425 \text{ cm}^2/s \times 483 \text{ cm}^{-3}}{31.2 \mu m}}{\frac{1.6 \times 10^{-19} C \times 19.425 \text{ cm}^2/s \times 483 \text{ cm}^{-3}}{31.2 \mu m} + \frac{1.6 \times 10^{-19} C \times 3.885 \text{ cm}^2/s \times 48.3 \text{ cm}^{-3}}{0.88 \mu m}}$$

$$= 58.5\%$$

The proportion of the electron current flowing through the space-charge region is 58.5%.

(d)

$$\frac{eD_n n_{p0}}{L_n} = \frac{en_i^2}{N_a} \sqrt{\frac{D_n}{\tau_n}}$$

$n_{p0} = \frac{n_i^2}{N_a}$
 $L_n = \sqrt{D_n \tau_n}$

As N_a decreases, the mobility μ_n increases, which leads to the increase of D_n .

So we can decrease the doping concentration of N_a in order to increase the electron current so that the proportion of electron current will increase too.